

```
//Experiment 6: Data Logging Using Arduino and Excel (Serial  
Communication)
```

```
#include <SD.h>
```

```
#include <SPI.h>
```

```
const int chipSelect = 10;
```

```
const int lm35Pin = A0;
```

```
void setup() {
```

```
    Serial.begin(9600);
```

```
    if (!SD.begin(chipSelect))
```

```
    {
```

```
        Serial.println("SD init failed!");
```

```
        while (1);
```

```
    }
```

```
    Serial.println("SD init done.");
```

```
}
```

```
void loop() {
```

```
    int adcValue = analogRead(lm35Pin);
```

```
    float temperatureC = (adcValue * 5.0 * 100.0) / 1024.0;
```

```
    File dataFile = SD.open("data.csv", FILE_WRITE);
```

```
    if (dataFile) {
```

```
    dataFile.print(millis() / 1000);  
    dataFile.print(",");  
    dataFile.println(temperatureC);  
    dataFile.close();  
    Serial.println("Data saved.");  
} else {  
    Serial.println("Error opening file!");  
}  
  
delay(1000); // Wait 1 second before next reading  
}
```